Artificial Neural Network

Supervised Machine Learning
Classification/Regression Algorithm

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Course Topics

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- ✓ What does ANN mean?
- ✓ How ANN works?
- ✓ Pros and Cons of ANN
- ✓ Applications of ANN
- ✓ Modeling ANN with Python

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What is ANN?

- Artificial Neural Networks are the biologically inspired simulations performed on the computer to perform certain specific tasks like clustering, classification, pattern recognition etc.

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How ANN Works? - ANN Layers

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- **Input layer** — It contains those units (artificial neurons) which receive input from the outside world on which network will learn, recognize about or otherwise process.
- **Output layer** — It contains units that respond to the information about how it's learned any task.
- **Hidden layer** — These units are in between input and output layers. The job of hidden layer is to transform the input into something that output unit can use in some way.

Most neural networks are fully connected that means to say each hidden neuron is fully connected to the every neuron in its previous layer(input) and to the next layer (output) layer.

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How ANN Works?



aX + b = Y

a = weight (w) , b = Bias (b)

a=1, b=1

1*10+1 = 11 (error :14)
1*5+1 = 6 (error :9)

a=2, b=2

2*10+2 = 22 (error :3)
2*5+2 = 12 (error :3)

a=3, b=3

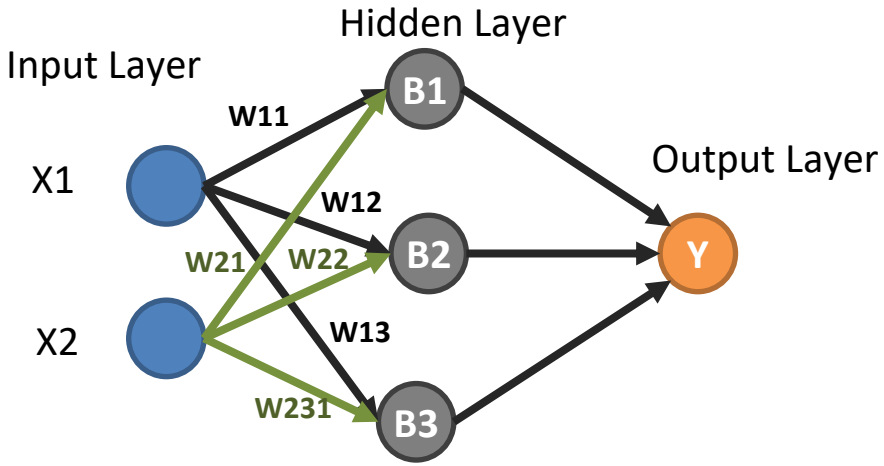
3*10+3 = 33 (error : -8)
3*5+3 = 18 (error :-3)

a=2, b=5

2*10+5 = 25 (error :0)
2*5+5 = 15 (error :0)

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How ANN works?



$$Y_{predict} = (X1*W11+X2*W21+B1) + (X1*W12+X2*W22+B2) + (X1*W13+X2*W23+B3)$$

$$error = Y_p*(1-Y_p)*(Y_a-Y_p)$$

$$W_{new} = W_{old} + l*(error)$$
$$B_{new} = B_{old} + l*(error)$$

ANN : Loan Approval

Input Layers
X1 = Credit Score
X2 = Salary Level

Output Layer
Y = Loan Approval

X1	X2	Y
0.8	0.6	1
0.3	0.7	0
0.6	0.4	0

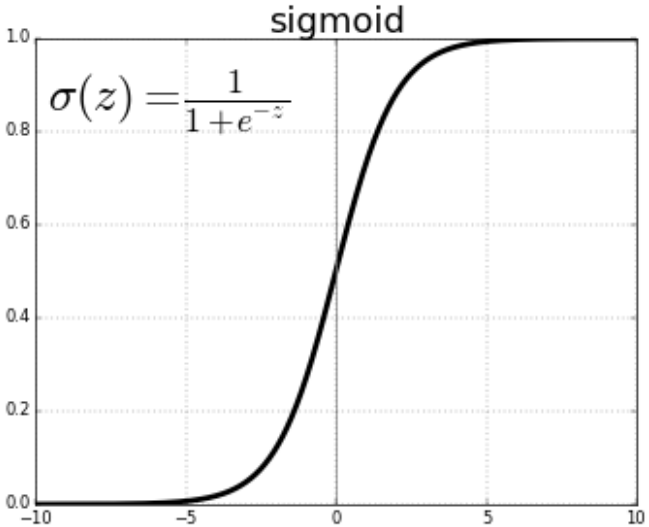
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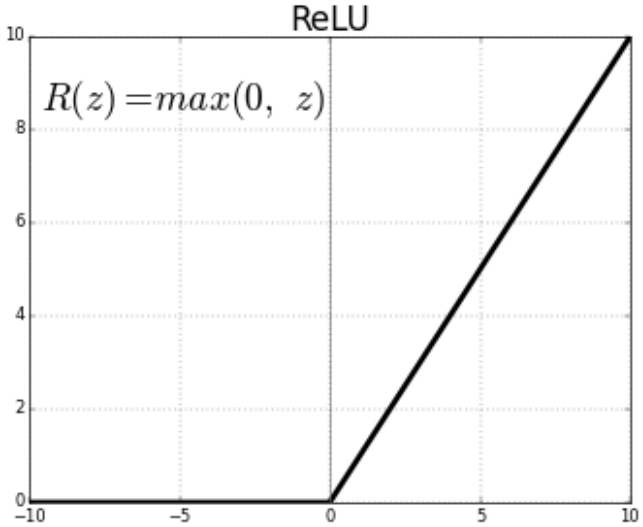
Stopping Criteria

- Max Iteration
- Validation score not improving
- Errors below threshold

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Hidden Layer Activation Functions





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Output Layer Activation Functions

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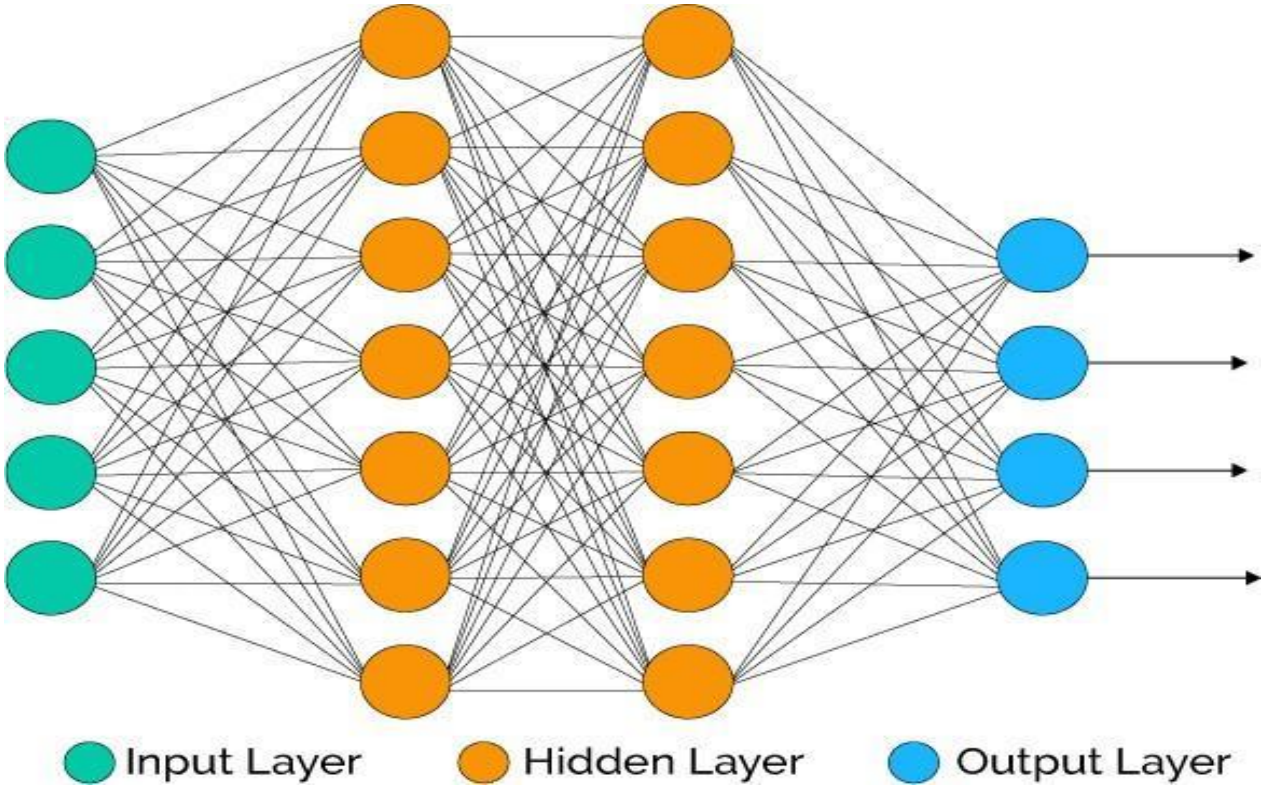
- Classification
  - Softmax
  - Sigmoid
- Regression
  - Linear

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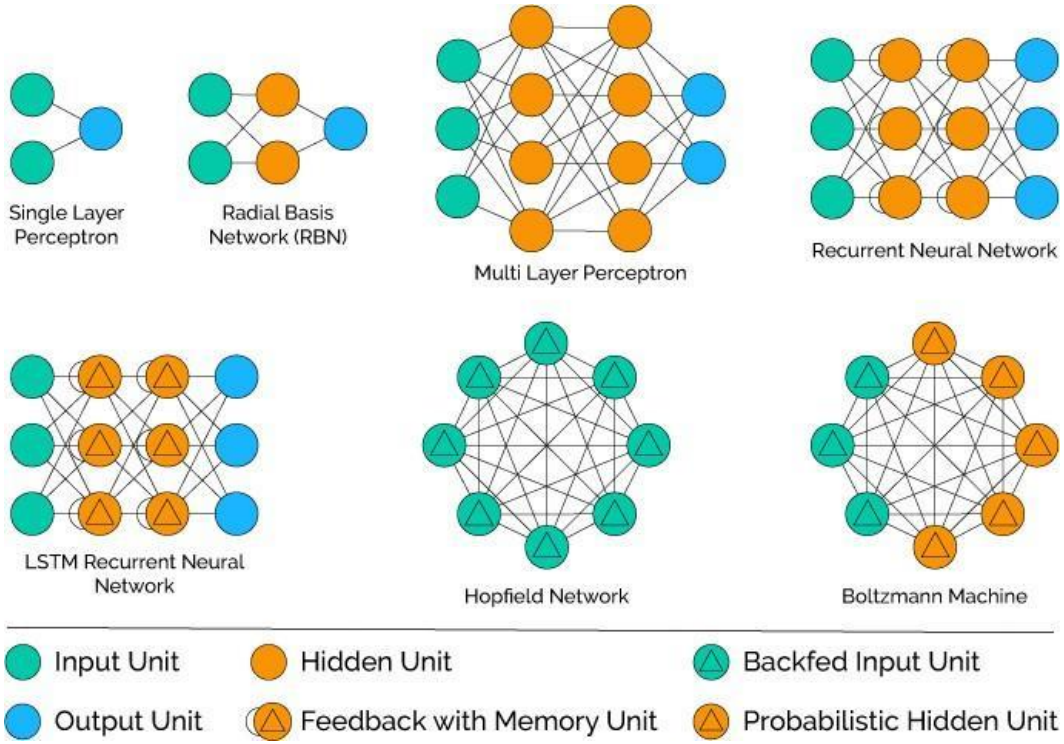
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How ANN Works? -Typical ANN



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How ANN Works? - Types of ANN



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Pros of ANN

- Good to model the non-linear data with large number of input features
- Widely used for high dimensionality data.
- ANNs can generalize — After learning from the initial inputs and their relationships
- ANN does not impose any restrictions on the input variables (like how they should be distributed)

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Cons of ANN

- NNs are useable only for numerical inputs, vectors with constant number of values, and datasets with non-missing data
- ANNs are computationally expensive.
- Black box, difficult to understand the modeling.
- Multi-layer neural networks are usually hard to train.
- Requires significant training data for model ANN compared to other ML algorithms.

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Applications of ANN

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- **Image Processing and Character recognition:** Given ANNs ability to take in a lot of inputs, process them to infer hidden as well as complex, non-linear relationships, ANNs are playing a big role in image and character recognition
- **Forecasting:** Forecasting is required extensively in everyday business decisions (e.g. sales, financial allocation between products, capacity utilization), in economic and monetary policy, in finance and stock market
- Game-playing and decision making (backgammon, chess, poker)
- **Medical Diagnosis:** Lung cancer, prostate cancer, colorectal cancer
- To accelerate reliability analysis of infrastructures subject to natural disasters

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MLP Classifier in action

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